

Introduction to Causal Inference for Software Engineering

A 90-minute hands-on tutorial

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Section 5

Causal Effect Estimation

What is the goal of estimation?

The goal is to compute a causal estimate — a precise quantity that answers our causal question — along with a measure of uncertainty from an estimand (the output of the identification phase).

Estimate	Question answered	Output	Example (SE)
Average Treatment Effect (ATE)	What is the average effect of the treatment across the whole population?	Single value + Confidence Interval (CI) + p-value	If all developers practiced pair programming, what would be the average improvement in code quality?
Average Treatment effect on the Treated (ATT)	What is the average effect among those who actually received the treatment?	Single value + Confidence Interval (CI) + p-value	Among developers who currently practice pair programming, how much does it help them?
Conditional Treatment Effect (CATE)	How does the effect vary across individuals or subgroups?	A function of covariates, one estimate per subgroup/individual	Does the effect vary by developer experience or task complexity?

What to take into account

- Before choosing a method, practical questions must be answered
 1. What is the form of the effect?
 - Is the relationship between treatment and outcome linear or non-linear? Assuming a linear model when reality is not, will lead to poor estimation quality.
 2. What type of data do you have?
 - Treatment, Outcome, Covariates (continuous, binary, ordinal...). Some estimation methods work only for specific types of data.

What to take into account

- Before choosing a method, practical questions must be answered
3. Do you need to transform the data?
 - No single answer here. Avoid transformations that change the identification results (e.g., feature engineering and dimensionality reduction). Can you interpret the results after transformation (e.g., what is the effect of a $\log(T)$ on Y)?
 4. How large is the adjustment set?
 - Few confounders: classical regression works well.
 - Many covariates (high-dimensional setting): use methods that separate the estimation of confounding from the estimation of the treatment effect (e.g., DoubleML).

Estimation techniques (1): Regression

- **Problem it solves:** How do I estimate the effect of the treatment on the outcome while controlling for confounders?
- **Key intuition:** Model the outcome as a function of both the treatment and the covariates. The treatment coefficient (or marginal effect) is your Average Treatment Effect (ATE).
 - Outcome = $f(\text{Treatment}, \text{Covariates}) \rightarrow$ read off the treatment coefficient
- **Main limitations:**
 - **Model misspecification:** You must correctly specify the model form. If the true relationship is non-linear and you use a linear model, the effect estimate is biased.
 - **Large adjustment sets:** When you have many covariates, classical regression struggles

Chapter 13 - Regression. “The Effect: An Introduction to Research Design and Causality”.

Nick Huntington-Klein. CRC Press <https://theeffectbook.net/index.html>

Estimation techniques (2): Matching

- **Problem it solves:** Given treated and untreated units that differ in their covariates (selection bias), how do I construct a comparable control group so that the outcome difference can be attributed to the treatment?
- **Key intuition:** Instead of modeling the outcome directly, matching “restructures” the data: each treated unit is paired with a similar untreated unit. After matching, the two groups are balanced and we can simply compare outcomes.
- **Main limitations:**
 - **Curse of dimensionality:** Exact matches become rare with many covariates
 - **Model dependence:** quality depends on how well the matching is estimated (e.g., propensity score needs to be estimated)
 - **Data loss:** Unmatched units are discarded → smaller effective sample

Stuart, Elizabeth A. "Matching methods for causal inference: A review and a look forward." (2010).

Chapter 14 - Matching. "The Effect: An Introduction to Research Design and Causality". Nick Huntington-Klein. CRC Press <https://theeffectbook.net/index.html>

6.4 - Propensity Scores and Inverse Probability Weighting (IPW). Introduction to Causal Inference. Brady Neals <https://youtu.be/YQDMINIQSA8?si=1Gyzc5HFUkcAmLKv>

Estimation techniques (3): Causal Machine Learning

- **Problem it solves:** How do I estimate treatment effects when the data is high-dimensional or the covariate–outcome relationship is complex and non-linear?
- **Key intuition:** Use flexible ML models to learn how covariates relate to treatment and outcome (nuisance functions), then isolate the treatment effect. Combining the flexibility of ML with valid causal inference.
- **Families of techniques:** Meta-Learners (S/T/X-Learner), Double ML, Causal Forests
- **Double ML in 3 steps:**
 1. Predict T from $X \rightarrow$ residuals $T^\sim$ (with a ML model)
 2. Predict Y from $X \rightarrow$ residuals $Y^\sim$ (with a ML model)
 3. Regress $Y^\sim$ on $T^\sim \rightarrow$ treatment effect (with a linear regression)
- **Main limitations:** Needs larger samples than regression; requires cross-fitting.

Chapter 6. Introduction to Causal Inference. Brady Neals. <https://www.bradyneal.com/slides/6%20-%20Estimation.pdf>
EconML <https://www.pywhy.org/EconML/spec/estimation.html>

Libraries and Reading Material

- EconML (python) <https://www.pywhy.org/EconML/index.html>
- DoubleML (python/R) <https://docs.doubleml.org/stable/index.html>
- CausalML (python) <https://causalml.readthedocs.io/en/latest/about.html>
- Matchit (R) <https://cran.r-project.org/web/packages/MatchIt/vignettes/MatchIt.html>
(Python: <https://github.com/jtuenner/pymatchit>)